

Seonghyun Park

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Summary

I'm a second-year Ph.D. student at [Graduate School of AI at KAIST](#), advised by [Sungsoo Ahn](#). Currently, my research focuses on **AI for Science (AI4Science)**, specifically on protein and small molecules. Recently, I led a project on machine learning Collective Variables (CVs) for enhanced sampling of proteins, by repurposing a foundation model.

Education

Ph.D. Korea Advanced Institute of Science and Technology (KAIST) , Kim Jaechul Graduate School of Artificial Intelligence • Structured and Probabilistic Machine Learning Lab @ Sungsoo Ahn • Interest: Protein-drug interactions, Molecular dynamics (MD)	Seoul, South Korea Feb 2025 – present
M.S. Pohang University of Science and Technology (POSTECH) , Graduate School of Computer Science and Engineering (CSE) • Machine Learning Lab @ Sungsoo Ahn • Interest: Graph Neural Networks (GNNs), Over-squashing	Pohang, South Korea Mar 2023 – Feb 2025
E.S. Institut National des Sciences Appliquées (INSA) Lyon , Bioinformatics Exchange Student	Lyon, France Jan 2022 – June 2022
B.S. Pohang University of Science and Technology (POSTECH) , Computer Science and Engineering (CSE)	Pohang, South Korea Feb 2019 – Feb 2023

Experience

Bagelcode , Business Analyst (BA) intern Game economy management and KPI analysis automation	Seoul, South Korea June 2021 – Aug 2021 3 months
Seller Hub , Product Manager (PM) intern Organization-wide task prioritization and landing funnel renewal	Seoul, South Korea July 2020 – Aug 2020 2 months

Publication

INDIBATOR: Diverse and Fact-Grounded Individuality for Multi-Agent Debate in Molecular Discovery Yunhui Jang, Seonghyun Park , Jaehyung Kim, Sungsoo Ahn 10.48550/arXiv.2602.01815 (Preprint)	2026
A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning Hyosoon Jang, Hyunjin Seo, Honghui Kim, Seonghyun Park , Taewon Kim, Yunhui Jang, Sungsoo Ahn 10.48550/arXiv.2602.13249 (Preprint)	2026
Riemannian MeanFlow Dongyeop Woo, Marta Skreta, Seonghyun Park , Kirill Neklyudov, Sungsoo Ahn 10.48550/arXiv.2602.07744 (ICML 2026)	July 2026

Learning Collective Variables from BioEmu with Time-lagged Generation Seonghyun Park , Kiyoun Seong, Soojung Yang, Rafael Gomez-Bombarelli, Sungsoo Ahn 10.48550/arXiv.2507.07390 (ICLR 2026)	Apr 2026
Transition Path Sampling with Improved Off-Policy Training of Diffusion Path Samplers Kiyoun Seong, Seonghyun Park , Seonghwan Kim, Woo Youn Kim, Sungsoo Ahn 10.48550/arXiv.2405.19961 (ICLR 2025)	Apr 2025
Non-backtracking Graph Neural Networks Seonghyun Park [†] , Narae Ryu [†] , Gahee Kim, Dongyeop Woo, Se-Young Yun [‡] , Sungsoo Ahn [‡] 10.48550/arXiv.2310.07430 (TMLR 2024, NeurIPS 2023 Workshop GLFrontiers Oral)	Sep 2024
Diffusion Probabilistic Models for Structured Node Classification Hyosoon Jang, Seonghyun Park , Sangwoo Mo, Sungsoo Ahn 10.48550/arXiv.2302.10506 (NeurIPS 2023)	Nov 2023